

9.3 Properties of Orthogonal Matrices



Core skills covered in this section:

- ▶ Orthogonal matrices: the inverse of the square matrix equals its transpose.
- ▶ Both the column vectors and row vectors of an orthogonal matrix form an orthonormal basis.
- ▶ Expanding the Gram matrix of an orthogonal matrix from different perspectives of matrix multiplication.
- ▶ The orthonormal basis of an orthogonal matrix preserves vector length and the angle between vectors.
- ▶ Orthogonal matrices make computing inverses and projections simple and numerically stable.
- ▶ The relationship between semi-orthogonal matrices and orthogonal matrices.
- ▶ The pseudo-inverse is a "generalized inverse" for non-square or non-invertible matrices.

This section introduces various properties of orthogonal matrices, and the content is quite "hardcore" – which is exactly why we laid the groundwork with the previous two sections first. Please make sure you have mastered the content of the previous two sections, especially understanding orthogonal matrices from a geometric point of view, before studying this section. This section draws on a range of linear algebra tools introduced earlier, such as the first and second perspectives of matrix multiplication, the Gram matrix, matrix inverses, matrix transpose, orthogonal projection, determinants, orthonormal bases, and block matrix multiplication.

Definition

A real $n \times n$ square matrix $\mathbf{A}$ is an orthogonal matrix if and only if

$$\mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbf{I}$$

Figure 1 shows the heatmap corresponding to $\mathbf{A}^T @ \mathbf{A} = \mathbf{I}$, $\mathbf{A}^T @ \mathbf{A}$ is the Gram matrix of the matrix $\mathbf{A}$.

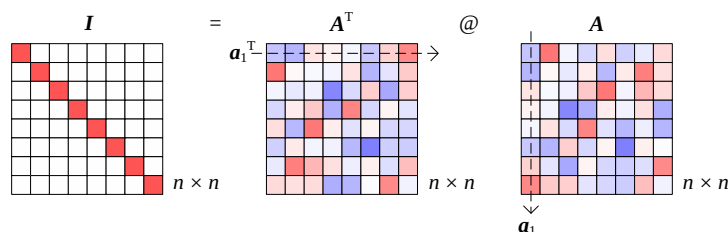


Figure 1. $\mathbf{A}^T @ \mathbf{A} = \mathbf{I}$

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also tells us that the inverse of $\mathbf{A}$ equals its transpose matrix

$$\mathbf{A}^{-1} = \mathbf{A}^T$$

This property of orthogonal matrices makes computing the inverse especially convenient.

Moreover, shows that $\mathbf{A}$ and $\mathbf{A}^T$ are both orthogonal matrices, as specifically shown in Figure 2. $\mathbf{A} @ \mathbf{A}^T$ is the Gram matrix of the matrix $\mathbf{A}^T$.

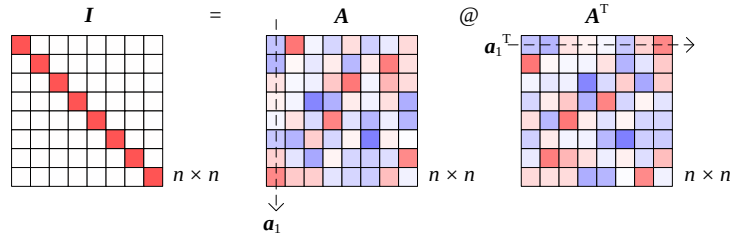


Figure 2. $\mathbf{A} @ \mathbf{A}^T = \mathbf{I}$

Below, let's use the different perspectives of matrix multiplication introduced earlier in this book to analyze the matrix multiplications corresponding to Figure 1 and Figure 2.

Expanding $\mathbf{A}^T @ \mathbf{A} = \mathbf{I}$

Let's first use the first perspective of matrix multiplication – the inner-product perspective – to expand $\mathbf{A}^T @ \mathbf{A} = \mathbf{I}$, i.e.,

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} @ \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^T @ \mathbf{a}_1 & \mathbf{a}_1^T @ \mathbf{a}_2 & \cdots & \mathbf{a}_1^T @ \mathbf{a}_n \\ \mathbf{a}_2^T @ \mathbf{a}_1 & \mathbf{a}_2^T @ \mathbf{a}_2 & \cdots & \mathbf{a}_2^T @ \mathbf{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_n^T @ \mathbf{a}_1 & \mathbf{a}_n^T @ \mathbf{a}_2 & \cdots & \mathbf{a}_n^T @ \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

We can also write in inner-product form (i.e., the inner-product perspective)

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{a}_1 & \mathbf{a}_1 \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_1 \cdot \mathbf{a}_n \\ \mathbf{a}_2 \cdot \mathbf{a}_1 & \mathbf{a}_2 \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_2 \cdot \mathbf{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_n \cdot \mathbf{a}_1 & \mathbf{a}_n \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_n \cdot \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

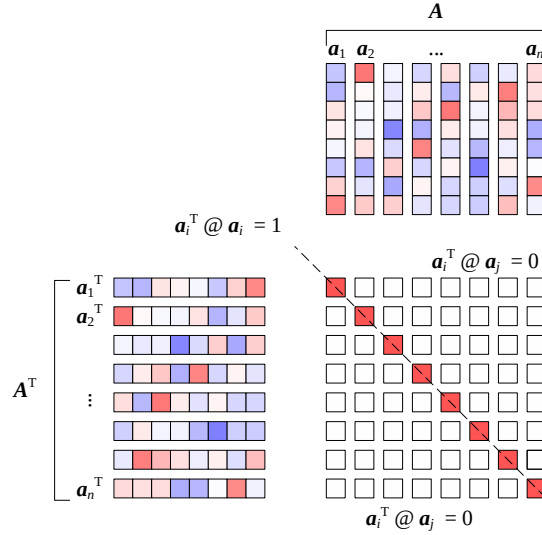
Figure 3. $A^T @ A = I$, the First Perspective of Matrix Multiplication (Inner-Product Perspective)

Figure 3's main-diagonal entries are 1, which shows that the column vectors of an orthogonal matrix are all unit vectors (direction vectors) — the vector length (magnitude, norm, L^2 norm, Euclidean norm) is 1, i.e.,

$$\mathbf{a}_i^T @ \mathbf{a}_i = \mathbf{a}_i \cdot \mathbf{a}_i = \langle \mathbf{a}_i, \mathbf{a}_i \rangle = \|\mathbf{a}_i\|_2^2 = 1$$

Figure 3's off-diagonal entries are 0, which shows that the column vectors of an orthogonal matrix are pairwise orthogonal, i.e.,

$$\mathbf{a}_i^T @ \mathbf{a}_j = \mathbf{a}_j^T @ \mathbf{a}_i = \mathbf{a}_i \cdot \mathbf{a}_j = \mathbf{a}_j \cdot \mathbf{a}_i = \langle \mathbf{a}_i, \mathbf{a}_j \rangle = \langle \mathbf{a}_j, \mathbf{a}_i \rangle = 0$$

where i is not equal to j .

In summary, the column vectors of the orthogonal matrix A are all unit vectors and are pairwise orthogonal.

Expanding $A^T @ A = I$

Now let's use the second perspective of matrix multiplication — the outer-product perspective — to expand $A^T @ A = I$, i.e.,

$$A^T A = \begin{bmatrix} \mathbf{a}^{(1)T} & \mathbf{a}^{(2)T} & \cdots & \mathbf{a}^{(n)T} \end{bmatrix} @ \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(n)} \end{bmatrix} = \mathbf{a}^{(1)T} @ \mathbf{a}^{(1)} + \mathbf{a}^{(2)T} @ \mathbf{a}^{(2)} + \cdots + \mathbf{a}^{(n)T} @ \mathbf{a}^{(n)} = I$$

Clearly, A 's row vectors are all nonzero vectors; therefore, $\mathbf{a}^{(j)T} @ \mathbf{a}^{(j)}$ is also rank one.

Moreover, $\mathbf{a}^{(j)T} @ \mathbf{a}^{(j)}$ is a symmetric matrix, i.e., it satisfies

$$\left(\mathbf{a}^{(i)} @ \mathbf{a}^{(i)T} \right)^T = \mathbf{a}^{(i)T} @ \mathbf{a}^{(i)}$$

As Figure 4 shows, the sum of n rank-one matrices is the identity matrix.

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Based on the content of the previous two sections of this chapter, Figure 4, every rank-one matrix in it, i.e., $\mathbf{a}^{(j)\top} @ \mathbf{a}^{(j)}$, is a projection matrix. $(\mathbf{a}^{(j)\top} @ \mathbf{a}^{(j)}) @ \mathbf{x}$ is the vector $\mathbf{x}$'s orthogonal projection onto $\mathbf{a}^{(j)\top}$.

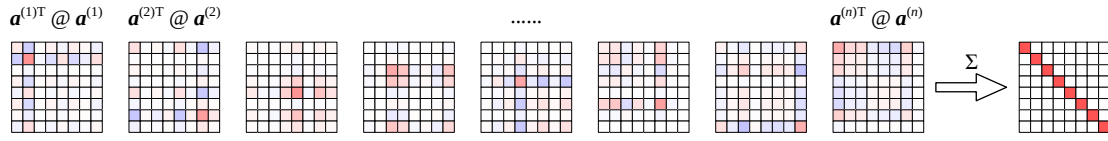


Figure 4. $\mathbf{A}^\top @ \mathbf{A} = \mathbf{I}$, the Second Perspective of Matrix Multiplication (Outer-Product Perspective)

Expanding $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$

For the matrix multiplication $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$, let's again first expand it using the first perspective of matrix multiplication – the inner-product perspective – i.e.,

$$\mathbf{A}\mathbf{A}^\top = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(n)} \end{bmatrix} @ \begin{bmatrix} \mathbf{a}^{(1)\top} & \mathbf{a}^{(2)\top} & \dots & \mathbf{a}^{(n)\top} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{(1)} @ \mathbf{a}^{(1)\top} & \mathbf{a}^{(1)} @ \mathbf{a}^{(2)\top} & \dots & \mathbf{a}^{(1)} @ \mathbf{a}^{(n)\top} \\ \mathbf{a}^{(2)} @ \mathbf{a}^{(1)\top} & \mathbf{a}^{(2)} @ \mathbf{a}^{(2)\top} & \dots & \mathbf{a}^{(2)} @ \mathbf{a}^{(n)\top} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}^{(n)} @ \mathbf{a}^{(1)\top} & \mathbf{a}^{(n)} @ \mathbf{a}^{(2)\top} & \dots & \mathbf{a}^{(n)} @ \mathbf{a}^{(n)\top} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

Writing in inner-product form (i.e., the inner-product perspective)

$$\mathbf{A}\mathbf{A}^\top = \begin{bmatrix} \mathbf{a}^{(1)} \cdot \mathbf{a}^{(1)} & \mathbf{a}^{(1)} \cdot \mathbf{a}^{(2)} & \dots & \mathbf{a}^{(1)} \cdot \mathbf{a}^{(n)} \\ \mathbf{a}^{(2)} \cdot \mathbf{a}^{(1)} & \mathbf{a}^{(2)} \cdot \mathbf{a}^{(2)} & \dots & \mathbf{a}^{(2)} \cdot \mathbf{a}^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}^{(n)} \cdot \mathbf{a}^{(1)} & \mathbf{a}^{(n)} \cdot \mathbf{a}^{(2)} & \dots & \mathbf{a}^{(n)} \cdot \mathbf{a}^{(n)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

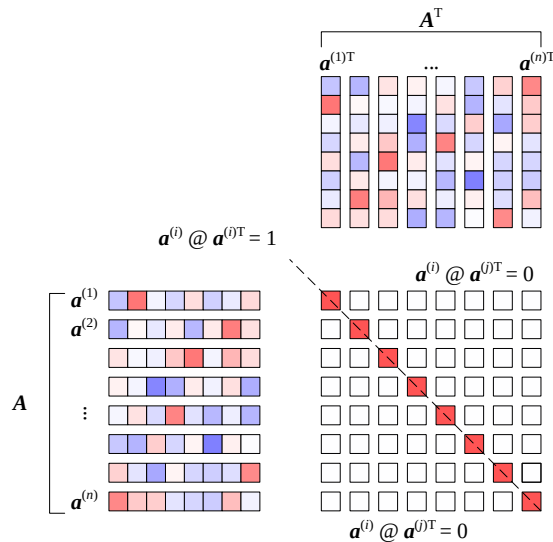


Figure 5. $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$, the First Perspective of Matrix Multiplication (Inner-Product Perspective)

Figure 5's main-diagonal entries are 1, which shows that the row vectors of the orthogonal matrix $\mathbf{A}$ are all unit vectors (direction vectors) – the vector length (magnitude, norm, L^2 norm, Euclidean norm) is 1, i.e.,

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$$\mathbf{a}^{(i)} @ \mathbf{a}^{(i)\top} = \mathbf{a}^{(i)} \cdot \mathbf{a}^{(i)} = \langle \mathbf{a}^{(i)}, \mathbf{a}^{(i)} \rangle = \|\mathbf{a}^{(i)}\|_2^2 = 1$$

Figure 5's off-diagonal entries are 0, which shows that the row vectors of an orthogonal matrix are pairwise orthogonal, i.e.,

$$\mathbf{a}^{(i)} @ \mathbf{a}^{(j)\top} = \mathbf{a}^{(j)} @ \mathbf{a}^{(i)\top} = \mathbf{a}^{(i)} \cdot \mathbf{a}^{(j)} = \mathbf{a}^{(j)} \cdot \mathbf{a}^{(i)} = \langle \mathbf{a}^{(i)}, \mathbf{a}^{(j)} \rangle = \langle \mathbf{a}^{(j)}, \mathbf{a}^{(i)} \rangle = 0$$

where i is not equal to j .

In summary, the row vectors of the orthogonal matrix $\mathbf{A}$ are all unit vectors and are pairwise orthogonal.

Expanding $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$

Expanding $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$ using the second perspective of matrix multiplication – the outer-product perspective – i.e.,

$$\mathbf{A}\mathbf{A}^\top = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n] @ \begin{bmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \vdots \\ \mathbf{a}_n^\top \end{bmatrix} = \mathbf{a}_1 @ \mathbf{a}_1^\top + \mathbf{a}_2 @ \mathbf{a}_2^\top + \cdots + \mathbf{a}_n @ \mathbf{a}_n^\top = \mathbf{I}$$

$\mathbf{A}$'s column vectors are all nonzero vectors; therefore, $\mathbf{a}_i @ \mathbf{a}_i^\top$ is rank one.

Moreover, $\mathbf{a}_i @ \mathbf{a}_i^\top$ is a symmetric matrix, satisfying

$$(\mathbf{a}_i @ \mathbf{a}_i^\top)^\top = \mathbf{a}_i @ \mathbf{a}_i^\top$$

As Figure 6 shows, the sum of n rank-one matrices is the identity matrix. Each rank-one matrix is also a projection matrix.

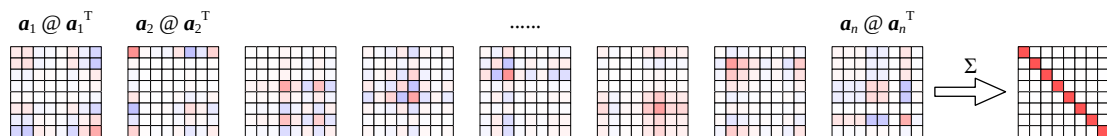


Figure 6. $\mathbf{A} @ \mathbf{A}^\top = \mathbf{I}$, the Second Perspective of Matrix Multiplication (Outer-Product Perspective)

Absolute Value of the Determinant Is 1: Area Is Preserved

If the matrix $\mathbf{A}$ is an orthogonal matrix, its determinant is ± 1 . The proof is also quite simple, as follows

$$\det(\mathbf{A})^2 = \det(\mathbf{A}^\top) \det(\mathbf{A}) = \det(\mathbf{A}^\top \mathbf{A}) = \det(\mathbf{I}) = 1$$

This book mentioned earlier that the identity matrix, rotation matrices, permutation matrices, reflection matrices, and combinations of them are all orthogonal matrices.

Clearly the identity matrix satisfies, i.e.,

$$\mathbf{I}^\top \mathbf{I} = \mathbf{I} \mathbf{I}^\top = \mathbf{I}$$

For example,

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T @ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The determinant of the identity matrix is 1.

The (about-the-origin) counterclockwise rotation matrix in the plane also satisfies

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}^T @ \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} @ \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The determinant of the rotation matrix in the equation above is 1.

Here's one way to understand it: swapping rows or columns of the identity matrix gives a permutation matrix. In matrix multiplication, a permutation matrix's role is to swap rows or columns. Permutation matrices are also orthogonal matrices, for example

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^T @ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} @ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The determinant of the permutation matrix in the equation above is -1.

⚠ Note that the determinant of a permutation matrix can be either -1 or 1, depending on whether the number of permutations (swaps) is odd or even. This also tells us that the determinant of an orthogonal matrix can be either 1 or -1.

Reflection matrices are also orthogonal matrices, for example

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}^T @ \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

⚠ Note that the determinant of a reflection matrix is also -1.

Orthonormal Basis

Since $\mathbf{A}$ and $\mathbf{A}^T$ are both orthogonal matrices, $\mathbf{A}$ and $\mathbf{A}^T$ are also both orthonormal bases.

First, $\mathbf{A}$'s column vectors, via linear combination, span $(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$, spanning $\mathbb{R}^n$.

Using the approach from the previous two sections, given an n-dimensional column vector $\mathbf{x}$, the following multiplication represents projecting $\mathbf{x}$ onto $\mathbf{A}$'s column vectors individually, then summing the results

$$\mathbf{I}\mathbf{x} = \mathbf{A}\mathbf{A}^T\mathbf{x} = (\mathbf{a}_1 @ \mathbf{a}_1^T + \mathbf{a}_2 @ \mathbf{a}_2^T + \dots + \mathbf{a}_n @ \mathbf{a}_n^T) \mathbf{x} = (\mathbf{a}_1 @ \mathbf{a}_1^T) @ \mathbf{x} + (\mathbf{a}_2 @ \mathbf{a}_2^T) @ \mathbf{x} + \dots + (\mathbf{a}_n @ \mathbf{a}_n^T) @ \mathbf{x}$$

Figure 6, every rank-one matrix in it, i.e., $\mathbf{a}_i @ \mathbf{a}_i^T$, is an orthogonal projection matrix.

$(\mathbf{a}_i @ \mathbf{a}_i^T) @ \mathbf{x}$ is the vector $\mathbf{x}$'s orthogonal projection onto $\mathbf{a}_i$.

Similarly, $\mathbf{A}^T$'s column vectors, via linear combination, span $(\mathbf{a}^{(1)T}, \mathbf{a}^{(2)T}, \dots, \mathbf{a}^{(n)T})$, also spanning $\mathbb{R}^n$. Please try to explain the following equation

$$\mathbf{I}\mathbf{x} = \mathbf{A}^T\mathbf{A}\mathbf{x} = (\mathbf{a}^{(1)T} @ \mathbf{a}^{(1)}) @ \mathbf{x} + (\mathbf{a}^{(2)T} @ \mathbf{a}^{(2)}) @ \mathbf{x} + \dots + (\mathbf{a}^{(n)T} @ \mathbf{a}^{(n)}) @ \mathbf{x}$$

Figure 4, every rank-one matrix in it, i.e., $\mathbf{a}^{(j)T} @ \mathbf{a}^{(j)}$, is also a projection matrix. $(\mathbf{a}^{(j)T} @ \mathbf{a}^{(j)}) @ \mathbf{x}$ is the vector $\mathbf{x}$'s orthogonal projection onto $\mathbf{a}^{(j)T}$.

Preserving Vector Length

For any n-dimensional vector $\mathbf{x}$, the magnitude of $\mathbf{Ax}$ equals the magnitude of $\mathbf{x}$, i.e.,

$$\|\mathbf{Ax}\| = \|\mathbf{x}\|$$

The proof is quite simple, as follows

$$\|\mathbf{Ax}\|^2 = (\mathbf{Ax})^T (\mathbf{Ax}) = \mathbf{x}^T \underset{I}{\mathbf{A}^T \mathbf{A}} \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2$$

Taking rotation as an example, as Figure 7 shows, from a geometric point of view, rotation does not change the length of a vector.

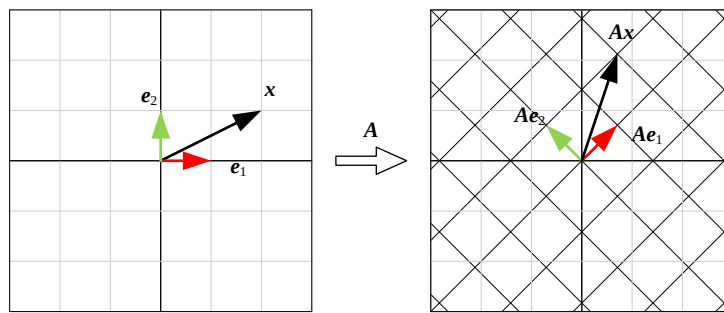


Figure 7. An Orthogonal Transformation Does Not Change Vector Length

For any n-dimensional vector, the magnitude of $\mathbf{A}^T \mathbf{x}$ equals the magnitude of $\mathbf{x}$, i.e.,

$$\|\mathbf{A}^T \mathbf{x}\| = \|\mathbf{x}\|$$



Please prove on your own, and explain what it means from a geometric point of view.

Preserving Angles and Inner Products

If $\mathbf{A}$ is an orthogonal matrix, then for two n-dimensional vectors $\mathbf{x}$ and $\mathbf{y}$, the angle between $\mathbf{Ax}$ and $\mathbf{Ay}$ equals the angle between $\mathbf{x}$ and $\mathbf{y}$.

The proof is also quite simple: we already know that the magnitude of $\mathbf{Ax}$ equals the magnitude of $\mathbf{x}$, and the magnitude of $\mathbf{Ay}$ equals the magnitude of $\mathbf{y}$. We only need to prove that the inner product of $\mathbf{Ax}$ and $\mathbf{Ay}$ equals the inner product of $\mathbf{x}$ and $\mathbf{y}$, i.e.,

$$(\mathbf{Ax}) \cdot (\mathbf{Ay}) = \mathbf{x} \cdot \mathbf{y}$$

Computing the inner product above using matrix multiplication

$$(\mathbf{Ax})^T @ (\mathbf{Ay}) = \mathbf{x}^T @ \mathbf{y}$$

Simplifying completes the proof

$$(\mathbf{Ax})^T @ (\mathbf{Ay}) = \mathbf{x}^T \underset{I}{\mathbf{A}^T \mathbf{A}} \mathbf{y} = \mathbf{x}^T \mathbf{y}$$

As Figure 8 shows, from a geometric point of view, an orthogonal transformation does not change the angle between vectors.

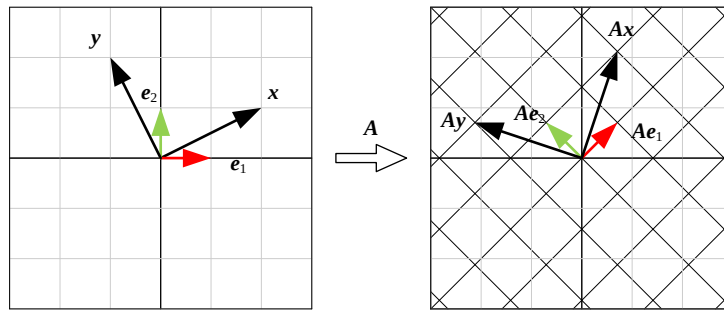


Figure 8. An Orthogonal Transformation Does Not Change the Angle Between Vectors

The Product of Orthogonal Matrices Is Still an Orthogonal Matrix

If A_1 and A_2 are both orthogonal matrices, then

$$(A_1 A_2)^T @ (A_1 A_2) = A_2^T A_1^T A_1 A_2 = A_2^T A_2 = I$$

Likewise

$$(A_1 A_2) @ (A_1 A_2)^T = A_1 A_2 A_2^T A_1^T = A_1 A_1^T = I$$

This means that composite geometric transformation matrices such as "rotation $\rightarrow$ reflection", "permutation $\rightarrow$ reflection", "rotation $\rightarrow$ permutation $\rightarrow$ reflection", and so on, are also all orthogonal matrices.

Semi-Orthogonal Matrices

A non-square matrix A_1 that satisfies

$$A_1^T A_1 = I$$

but does not satisfy

$$A_1 A_1^T = I$$

is called semi-orthogonal (semi-orthogonal matrix). The column vectors of this matrix also form an orthonormal basis.

In fact, Figure 1, the matrix A – with any number of its columns removed, the remaining matrix is a semi-orthogonal matrix A_1 , specifically as Figure 9 shows.

⚠ Note that the removed column vectors need not be consecutive.

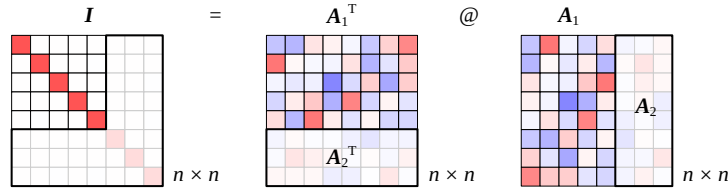


Figure 9. Removing Some Columns from an Orthogonal Matrix Gives a Semi-Orthogonal Matrix

As Figure 10 shows, $\mathbf{A}$ matrix's removed portion ($\mathbf{A}_2$) and its retained submatrix ($\mathbf{A}_1$) span spaces that are orthogonal complements of each other.

Writing $\mathbf{A}$ as $[\mathbf{A}_1, \mathbf{A}_2]$, and using the block matrix multiplication introduced earlier in this book, the matrix multiplication $\mathbf{A}^T \mathbf{A}$ can be expanded as

$$\begin{aligned} \mathbf{A}^T \mathbf{A} &= [\mathbf{A}_1^T \quad \mathbf{A}_2^T] [\mathbf{A}_1 \quad \mathbf{A}_2] \\ &= \begin{bmatrix} \mathbf{A}_1^T \\ \mathbf{A}_2^T \end{bmatrix} [\mathbf{A}_1 \quad \mathbf{A}_2] = \begin{bmatrix} \mathbf{A}_1^T \mathbf{A}_1 & \mathbf{A}_1^T \mathbf{A}_2 \\ \mathbf{A}_2^T \mathbf{A}_1 & \mathbf{A}_2^T \mathbf{A}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \end{aligned}$$

In the equation above, $\mathbf{A}_1^T \mathbf{A}_1$ and $\mathbf{A}_2^T \mathbf{A}_2$ are both Gram matrices of semi-orthogonal matrices.

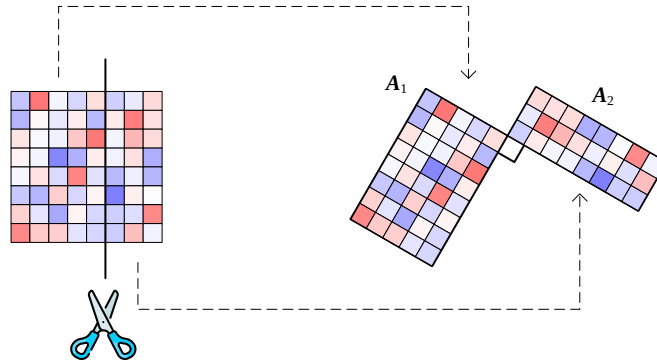


Figure 10. The Spanned Spaces Are Orthogonal Complements of Each Other

For example, given the 3×2 matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Now let's compute $\mathbf{A}^T \mathbf{A}$

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The result is the identity matrix. This means that $\mathbf{A}$'s column vectors are all unit vectors and are pairwise orthogonal.

Now let's compute $\mathbf{A} \mathbf{A}^T$

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$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This result is not the identity matrix.

Therefore, $\mathbf{A}$ is a semi-orthogonal matrix — $\mathbf{A}$'s column vectors form an orthonormal basis; however, $\mathbf{A}$ is not an orthogonal matrix.

Also, please note that if matrix $\mathbf{A}$ satisfies $\mathbf{A}\mathbf{A}^T = \mathbf{I}$ but does not satisfy $\mathbf{A}^T\mathbf{A} = \mathbf{I}$, this means matrix $\mathbf{A}$'s row vectors form an orthonormal basis. Sometimes, we also call this matrix $\mathbf{A}$ a semi-orthogonal matrix as well.

For example, consider the 2×3 matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 4/5 & 0 & 3/5 \\ -3/5 & 0 & 4/5 \end{bmatrix}$$

Now let's compute $\mathbf{A}^T\mathbf{A}$

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 4/5 & -3/5 \\ 0 & 0 \\ 3/5 & 4/5 \end{bmatrix} \begin{bmatrix} 4/5 & 0 & 3/5 \\ -3/5 & 0 & 4/5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The result is not the identity matrix.

Now let's compute $\mathbf{A}\mathbf{A}^T$

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 4/5 & 0 & 3/5 \\ -3/5 & 0 & 4/5 \end{bmatrix} \begin{bmatrix} 4/5 & -3/5 \\ 0 & 0 \\ 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This result is the identity matrix.

This means that $\mathbf{A}$'s row vectors are all unit vectors and are pairwise orthogonal.

Therefore, $\mathbf{A}$ can also be called a semi-orthogonal matrix — $\mathbf{A}$'s row vectors form an orthonormal basis; however, $\mathbf{A}$ is not an orthogonal matrix.

Orthogonal Projection for a General Matrix

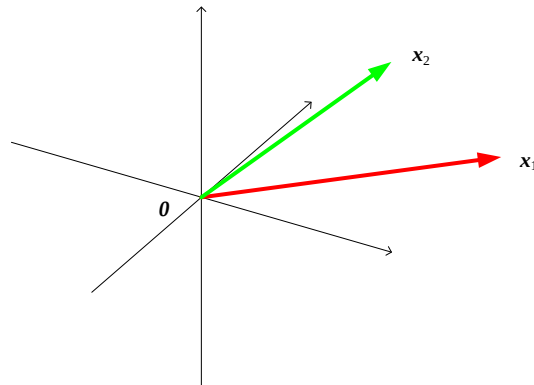
We are so "eager" for orthogonal matrices because, when dealing with orthogonal projection problems, computations with a general matrix are quite troublesome!

Let's look at an example below.

Given the following two three-dimensional column vectors

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

As Figure 11 shows, clearly $\mathbf{x}_1$ and $\mathbf{x}_2$ are not collinear; the non-orthogonal basis $[\mathbf{x}_1, \mathbf{x}_2]$ spans the plane $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$ in three-dimensional space.

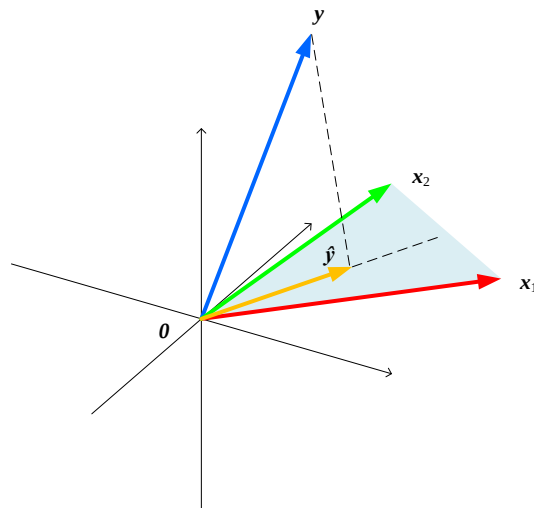
Figure 11. Specific Positions of x_1 and x_2

Let's also give a three-dimensional column vector y

$$y = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

As Figure 12 shows, the vector y is not on the plane $\text{span}(x_1, x_2)$.

The vector y 's orthogonal projection onto $\text{span}(x_1, x_2)$ is $\hat{y}$. $\hat{y}$ is clearly on the $\text{span}(x_1, x_2)$ plane!

Figure 12. y Projected Onto the $\text{span}(x_1, x_2)$ Plane

Since $[x_1, x_2]$ is a non-orthogonal basis for $\text{span}(x_1, x_2)$; that is, $\hat{y}$ can be written as a linear combination of x_1 and x_2

$$\hat{\mathbf{y}} = b_1 \mathbf{x}_1 + b_2 \mathbf{x}_2 = b_1 \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + b_2 \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

To find $\hat{\mathbf{y}}$, we need to compute b_1 and b_2 .

Writing this in matrix multiplication form

$$\hat{\mathbf{y}} = \underbrace{\begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}}_{\mathbf{b}} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \mathbf{b} = \begin{bmatrix} 2 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{b} = \mathbf{X}\mathbf{b}$$

The matrix $\mathbf{X}$ is not square, so it is clearly not invertible!

Let's think of another way!

As Figure 13 shows, the vector $\mathbf{y}$ can be orthogonally decomposed into two parts:

- a) $\hat{\mathbf{y}}$, on the $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$ plane;
- b) $\boldsymbol{\varepsilon}$, perpendicular to $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$.

$\boldsymbol{\varepsilon}$ is the difference between $\mathbf{y}$ and $\hat{\mathbf{y}}$

$$\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\mathbf{b}$$

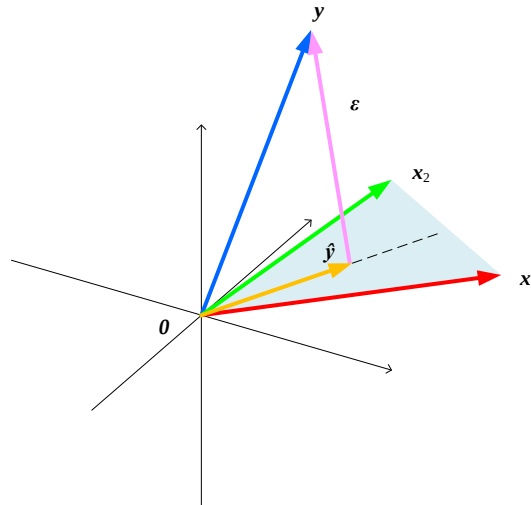


Figure 13. $\mathbf{y}$ Orthogonally Decomposed Into Two Parts

Let's start with $\boldsymbol{\varepsilon}$!

Since $\boldsymbol{\varepsilon}$ is perpendicular to $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$, $\boldsymbol{\varepsilon}$ must necessarily be perpendicular to $\mathbf{x}_1$ and $\mathbf{x}_2$ individually, so we have

$$\begin{aligned} \mathbf{x}_1^T @ \boldsymbol{\varepsilon} &= 0 \\ \mathbf{x}_2^T @ \boldsymbol{\varepsilon} &= 0 \end{aligned}$$

Combining these two equations

$$\begin{bmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \end{bmatrix} @ \boldsymbol{\varepsilon} = \mathbf{0} \Rightarrow \mathbf{X}^\top \boldsymbol{\varepsilon} = \mathbf{0}$$

Substituting the previous two equations together gives

$$\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$$

Rearranging the equation above gives

$$\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \mathbf{b}$$

The equation above is actually the result of left-multiplying both sides of the equation $\mathbf{y} = \mathbf{X}\mathbf{b}$ by $\mathbf{X}^\top$.

Since $\mathbf{X}$ has full column rank (its column vectors are not collinear), the Gram matrix $\mathbf{X}^\top \mathbf{X}$ is invertible; therefore

$$\mathbf{b} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

In fact, as Figure 14 shows, $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is called the pseudo-inverse, also known as the generalized inverse.

⚠ Note that this book mentioned earlier that matrix inverses have the property $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$; this equation holds only under the premise that $\mathbf{A}$ and $\mathbf{B}$ are both invertible. In the equation above, the matrix $\mathbf{X}$ is generally a tall, narrow matrix and is clearly not invertible.

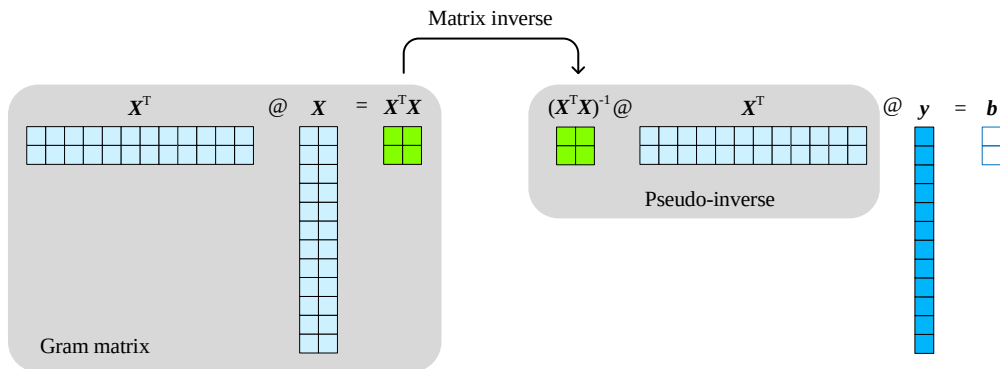


Figure 14. Computing $\mathbf{b}$

Simply put, the pseudo-inverse is a "generalized inverse" for a non-square matrix (such as $\mathbf{X}$) or a non-invertible matrix, used to approximate a solution when the inverse cannot be computed directly.

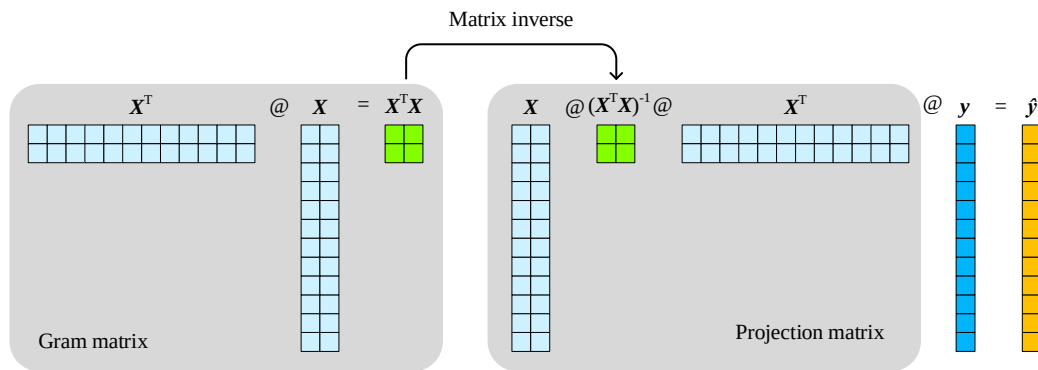
⚠ One point must be emphasized: only when $\mathbf{X}$ has full column rank does $\mathbf{X}^\top \mathbf{X}$ have an inverse.

In this way, we obtain the result that $\mathbf{y}$'s projection onto the $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$ plane, $\hat{\mathbf{y}}$, can be written as

$$\hat{\mathbf{y}} = \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

where $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is the projection matrix for orthogonally projecting $\hat{\mathbf{y}}$ onto the vector space constructed from $\mathbf{X}$'s columns.

In the equation above, $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is often called the hat matrix.

Figure 15. Computing $\hat{y}$

$X(X^T X)^{-1} X^T$ is an idempotent matrix.

This book mentioned earlier that a projection matrix is also an idempotent matrix; multiplying an idempotent matrix by itself still gives itself, i.e.,

$$\left(X (X^T X)^{-1} X^T \right)^2 = X \underbrace{(X^T X)^{-1} X^T X}_{I} (X^T X)^{-1} X^T = X (X^T X)^{-1} X^T$$

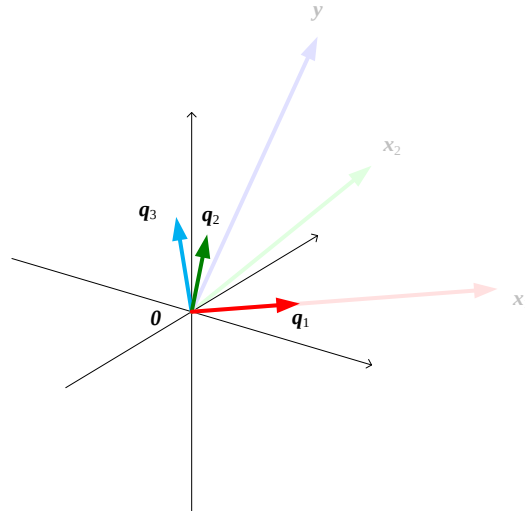
Orthogonal Projection Using a Semi-Orthogonal Matrix

If we can find an orthonormal basis on the $\text{span}(x_1, x_2)$ plane that also spans $\text{span}(x_1, x_2)$ plane, then carrying out the projection computation above becomes "as easy as pie"!

Through a certain special method, we have indeed found an orthonormal basis

$$Q_1 = [q_1 \quad q_2] = \begin{bmatrix} \sqrt{2}/2 & \sqrt{6}/6 \\ \sqrt{2}/2 & -\sqrt{6}/6 \\ 0 & \sqrt{6}/3 \end{bmatrix}$$

As Figure 16 shows, the column vectors q_1 and q_2 span the plane $\text{span}(q_1, q_2)$, which is equivalent to $\text{span}(x_1, x_2)$.

Figure 16. Orthonormal Basis $[q_1, q_2, q_3]$

? $\text{span}(q_1, q_2)$ – why is this equivalent to $\text{span}(x_1, x_2)$? How do we compute q_1 and q_2 ? These questions will be answered in the next section.

The column vectors q_1 and q_2 both lie on the $\text{span}(x_1, x_2)$ plane; q_1 and q_2 are both unit vectors, and they are orthogonal. Therefore, $[q_1, q_2]$ is an orthonormal basis.

This also tells us that Q_1 is a semi-orthogonal matrix, i.e., it satisfies $Q_1^T Q_1 = I$.

Take Q_1 and substitute it into the equation above in place of X , giving us

$$\begin{aligned}\hat{y} &= Q_1 \underbrace{(Q_1^T Q_1)^{-1}}_I Q_1^T y \\ &= Q_1 Q_1^T y\end{aligned}$$

Writing Q_1 as $[q_1, q_2]$, and expanding the matrix multiplication in the equation above

$$\begin{aligned}\hat{y} &= Q_1 Q_1^T y \\ &= [q_1 \quad q_2] \begin{bmatrix} q_1^T \\ q_2^T \end{bmatrix} y \\ &= (q_1 @ q_1^T + q_2 @ q_2^T) y \\ &= q_1 @ q_1^T y + q_2 @ q_2^T y\end{aligned}$$

This is equivalent to orthogonally projecting y onto q_1 and q_2 individually, then combining the results!

The semi-orthogonal matrix Q_1 's property $Q_1^T Q_1 = I$ plays a crucial role here!

Careful readers may have already noticed that spanning three-dimensional space requires three linearly independent "sticks."

An orthonormal basis that spans three-dimensional space, however, requires the three "sticks" to each have length 1 and be pairwise orthogonal.

Now that we have the orthogonal unit vectors q_1 and q_2 , we still need one more unit vector, orthogonal to both q_1 and q_2 ; let's call it q_3 .

And q_3 can be solved for using the equation below

$$\mathbf{q}_3 @ \mathbf{q}_3^T = \mathbf{I} - \mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T$$

Clearly, $\mathbf{q}_3$ is the normal vector of the $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$ plane; this also means that $\mathbf{q}_3$, as Figure 13 shows, is parallel to $\mathbf{e}$.



You'll soon discover that this example actually lays the groundwork for three topics: Gram-Schmidt orthogonalization, QR decomposition, and linear regression! Gram-Schmidt orthogonalization is precisely the method for obtaining $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ in this example!

How Do We Obtain an Orthogonal Matrix?

Since orthogonal matrices (and even semi-orthogonal matrices) have so many "excellent" properties, how do we go about obtaining an orthogonal matrix?

Later in this book, QR decomposition, spectral decomposition, and singular value decomposition will all be introduced, each of which produces an orthogonal matrix.



Note that you don't need to master these matrix decompositions here — this book will explain them in detail later.

Using QR decomposition, any real matrix $\mathbf{A}$ can be decomposed as

$$\mathbf{A} = \mathbf{Q}\mathbf{R}$$

where $\mathbf{Q}$ is an orthogonal matrix.



Note that in some cases $\mathbf{Q}$ is a semi-orthogonal matrix.



The Gram-Schmidt orthogonalization introduced first in the next section is, in essence, the process of converting $\mathbf{A}$'s column vectors into $\mathbf{Q}$'s column vectors.

Using spectral decomposition, a symmetric matrix $\mathbf{A}$ can be decomposed as

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

where $\mathbf{V}$ is an orthogonal matrix.

Using singular value decomposition, any real matrix $\mathbf{A}$ can be decomposed as

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

where $\mathbf{U}$ and $\mathbf{V}$ are both orthogonal matrices.



Note that in some cases $\mathbf{U}$ and $\mathbf{V}$ can also be semi-orthogonal matrices.

These three matrix decompositions are all topics that will be covered in depth later in this book; clearly, orthogonal matrices play an important role in them. This is exactly why we devoted three whole sections to introducing orthogonal matrices.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Each of the following matrices $\mathbf{A}$ is an orthogonal matrix. Please expand $\mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbf{I}$ using both the first and second perspectives of matrix multiplication.

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▶ $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Q2. Each of the following matrices A is a semi-orthogonal matrix. Please expand $A^T A = I$, $AA^T = I$ using both the first and second perspectives of matrix multiplication.

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q3. Given the following three column vectors, please compute the coordinates of $\mathbf{a}_3$'s orthogonal projection onto $\text{span}(\mathbf{a}_1, \mathbf{a}_2)$.

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Q4. Please use the pseudo-inverse to solve the following system of linear equations.

▶ $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$